SOMASHEKHAR JAVOORU

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Summary

Software engineering candidate and 2026 ECE graduate with hands-on experience building high-performance C, C++, and Python applications. Strong foundation in Data Structures, Algorithms, and Object-Oriented Programming, combined with practical expertise in Linux environments, networking protocols (TCP/IP, HTTP, MQTT), and concurrent execution. Proven track record of debugging complex systems, executing rigorous unit tests, and leveraging Git for version control to deliver reliable software solutions.

Education

Bapuji Institute of Engineering and Technology (Davanagere) *Aug 2022 – Jun 2026*
Bachelor of Engineering in Electronics and Communication | **CGPA: 8.75/10.0**
Coursework: Data Structures & Algorithms, Object-Oriented Programming, Networking Concepts, Operating Systems, Linux Internals

SPANDANA PU SCIENCE COLL, KULKARNIPET, ILKAL *Pre-University Course (PUC)*
Science | Karnataka Department of Pre-University Education | **86.6%**

Technical Skills

- **Programming Languages:** C, C++, Python, Java
- **Computer Science Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Concurrency, Memory Management
- **Operating Systems & Networking:** Linux, TCP/IP, HTTP, MQTT, Wi-Fi, L2/L3 Concepts, RTOS
- **Software Engineering & Testing:** Git, Unit Testing, Debugging, Root-Cause Analysis, Software Integration, Agile SDLC
- **AI & Development Tools:** Machine Learning (CNN pipelines), Computer Vision, Data Telemetry, VS Code

Experience

Dextris Infoservices Pvt Ltd *Jun 2026 – Present*
Embedded Engineer Intern
- Engineered modular C/C++ applications using Object-Oriented Programming (OOP) principles, optimizing execution algorithms to decrease software response time by 25% for high-speed data telemetry.
- Built an HTTP-based video streaming subsystem utilizing Python and C++, achieving stable 30 FPS transmission across wireless networks for real-time monitoring.
- Spearheaded root-cause analysis and hardware-software debugging during integration, reducing application boot-up latency by 1.5 seconds.
- Managed continuous software delivery using Git version control and coordinated remote Over-The-Air (OTA) software updates.
- Optimized concurrent task execution and thread scheduling using FreeRTOS, reducing CPU idle time by 20% to improve overall resource efficiency.

EchoBrains *Jan 2026 – May 2026*
Embedded Systems Intern
- Developed and optimized C/C++ control algorithms, achieving <1 ms execution latency for time-critical software applications.
- Conducted comprehensive unit testing and verification on continuous data streams, ensuring 0% packet loss across concurrent communication buses.

Projects

Real-Time Face Recognition System | *C/C++, Python, HTTP, Wi-Fi* *Jun 2026* | Link
- Architected C/C++ software to interface with camera hardware, executing computer vision algorithms with <200 ms processing latency.
- Synchronized concurrent networking routines for local display handling and HTTP data transmission via Wi-Fi to a centralized server.
- Resolved critical memory leaks during continuous streaming by applying advanced memory management and root-cause debugging techniques.

AI-Powered Agriculture Robot | *Python, C/C++, Linux, Machine Learning* *Dec 2025* | Link
- Built Python-based CNN image processing pipelines within a Linux environment to autonomously classify conditions, improving algorithmic routing efficiency by 15%.
- Developed C-based algorithms utilizing precise timing logic for motor control, improving automated inspection accuracy by 20%.

Automated Gas Leakage Detection System | *C/C++, MQTT, Networking* *Mar 2025* | Link
- Deployed MQTT telemetry streaming over Wi-Fi to transmit real-time sensor data for remote monitoring, achieving 99.9% system uptime.
- Programmed event-driven automated software interrupts (ISRs) to trigger safety protocols upon hazard detection in <500 ms.

Certifications & Awards

- **Smart India Hackathon:** Engineered hardware-accelerated IoT and networking solutions with a 4-member cross-functional team.
- **Dextris Recognition:** Awarded for core contributions to software architecture, automated OTA deployment, and software bring-up.
- **HackerRank Java (Basic):** Certified in core programming concepts, data structures, and object-oriented logic.